

Wi-fi History

~ Wi-fi6



Table of Contents

I ~ 802.11g

II Wi-fi 4

III Wi-fi 5

IV Wi-fi 6

I ~ 802.11g

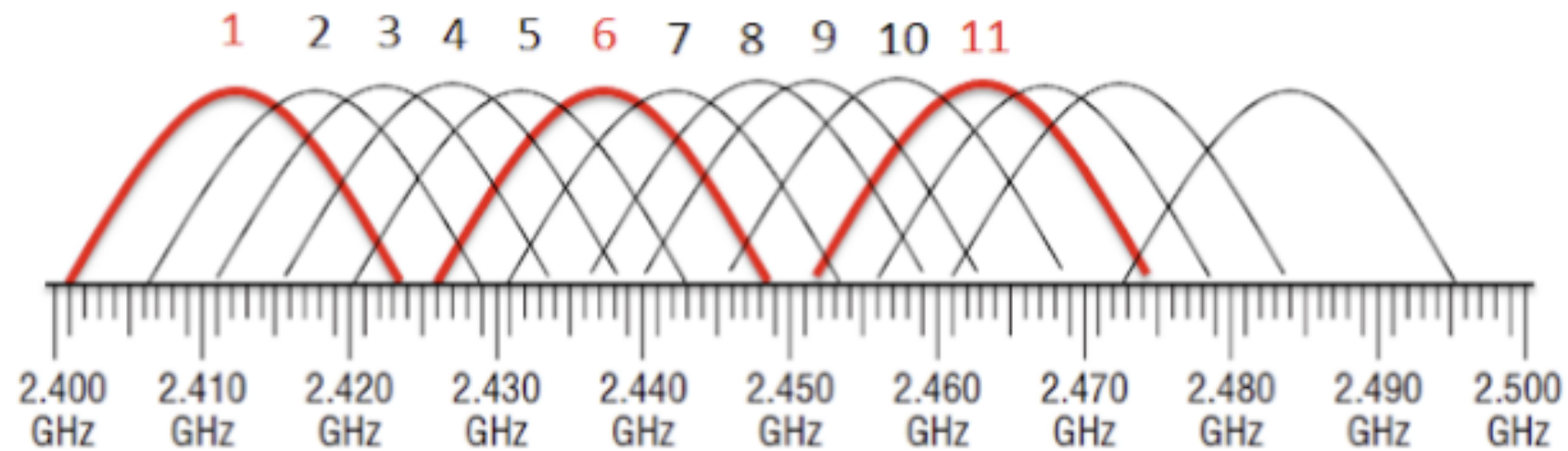
Standard	802.11	802.11b	802.11a	802.11g
Year	1997	1999	1999	2003
Frequency band	2.4GHz	2.4GHz	5GHz	2.4GHz
channel bandwidth	20MHz	20MHz	20MHz	20MHz
Max data rate	1~2 Mbps	11 Mbps	54 Mbps	54 Mbps

II wi-fi 4

Standard	802.11n	802.11ac	802.11ax
Year	2009	2013	2019
Frequency band	2.4GHz, 5GHz	5GHz	2.4GHz, 5GHz
channel bandwidth	20, 40	20, 40, 80, 160	20, 40, 80, 160
Max data rate	600 Mbps	7Gbps	9.6 Gbps
MIMO	MIMO (up to 4X4)	MU-MIMO (DL, up to 8X8)	MU-MIMO (DL&UL)
Modulation	OFDM	OFDM	OFDM, OFDMA
Data subcarrier modulation	BPSK, QPSK, 16QAM, 64QAM	BPSK, QPSK, 16QAM, 64QAM, 256QAM	BPSK, QPSK, 16QAM, 64QAM, 256QAM, 1024QAM

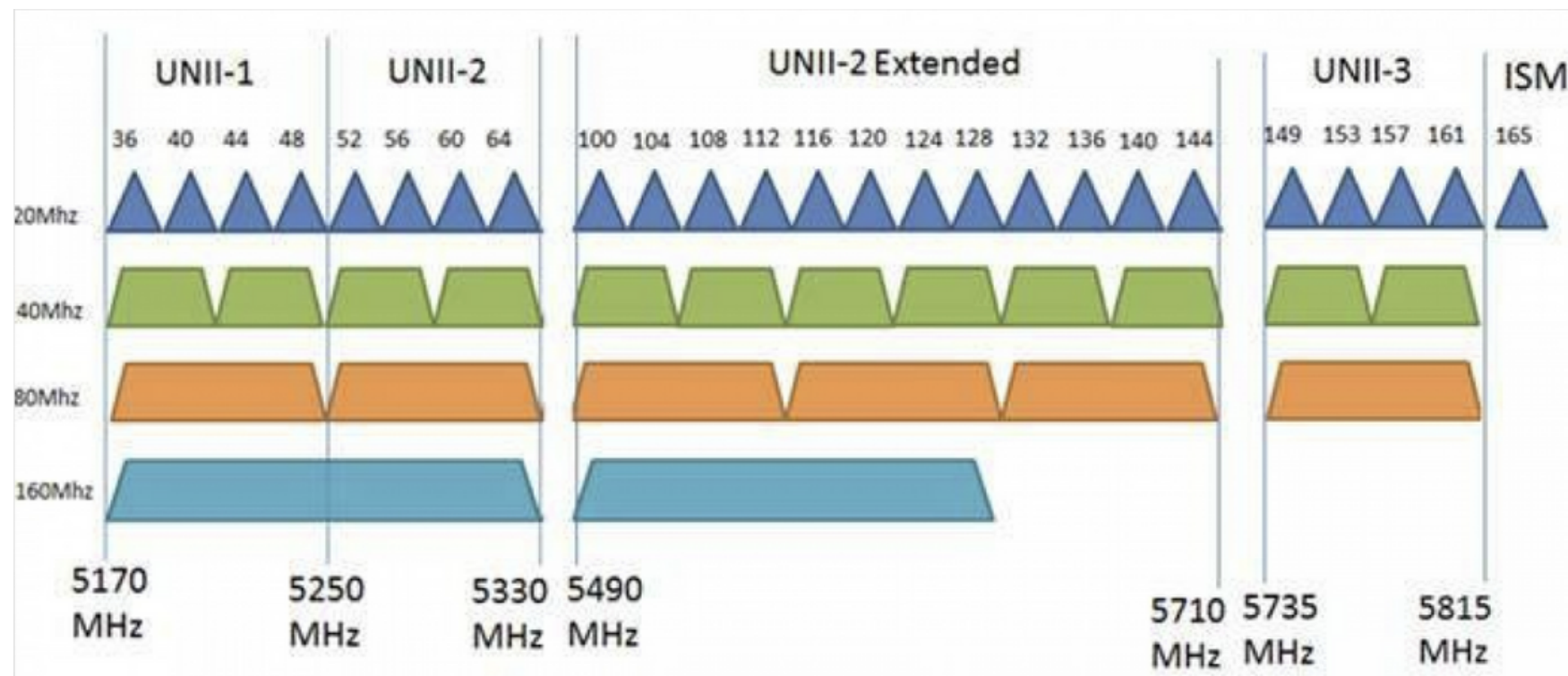
II wi-fi 4

Channel Bandwidth



2.4 GHz

- slower speed
- longer range
- more radio interference



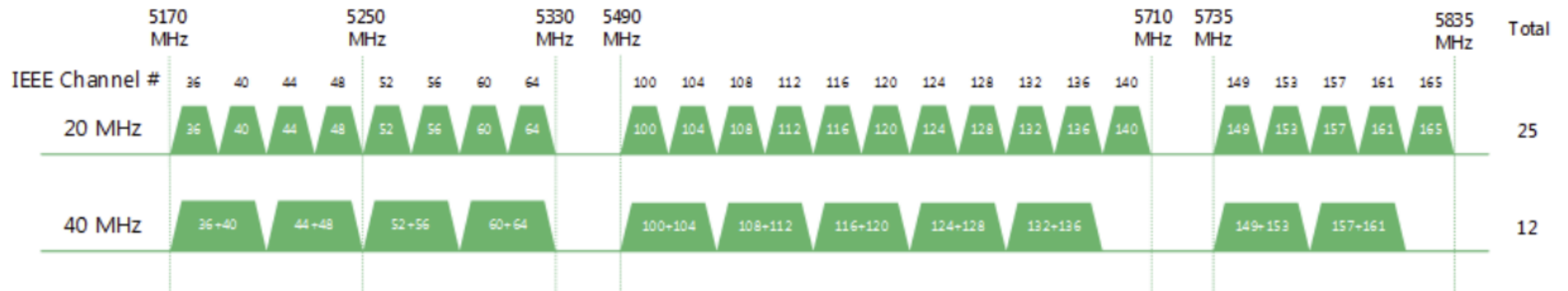
5GHz

- faster speed
- shorter range
- less radio interference

II wi-fi 4

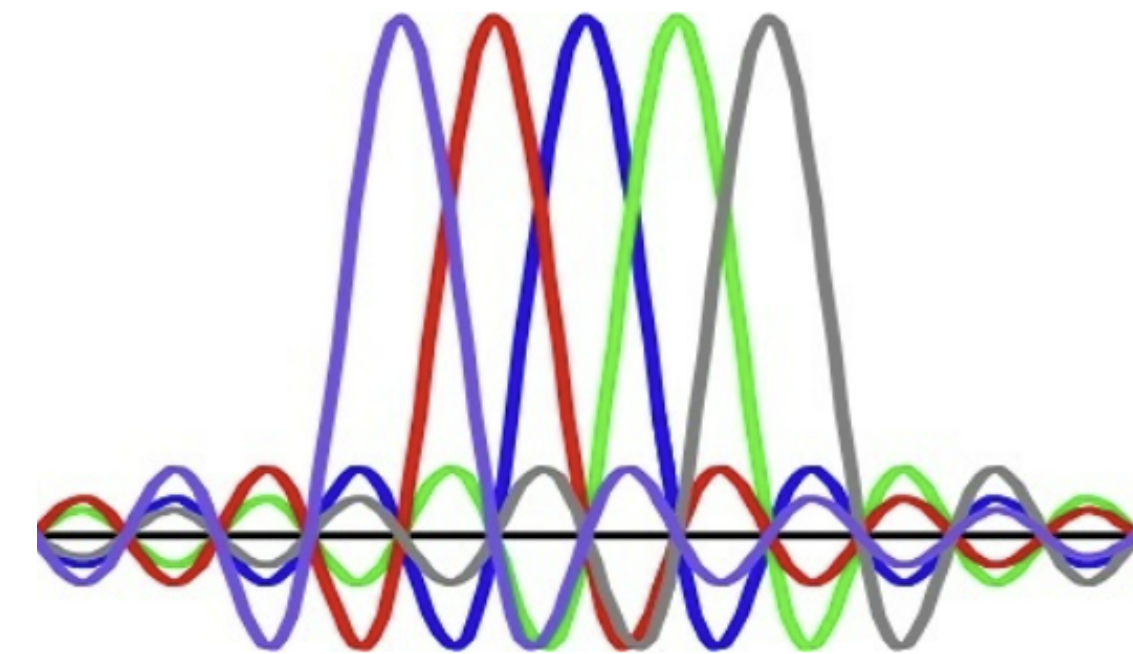
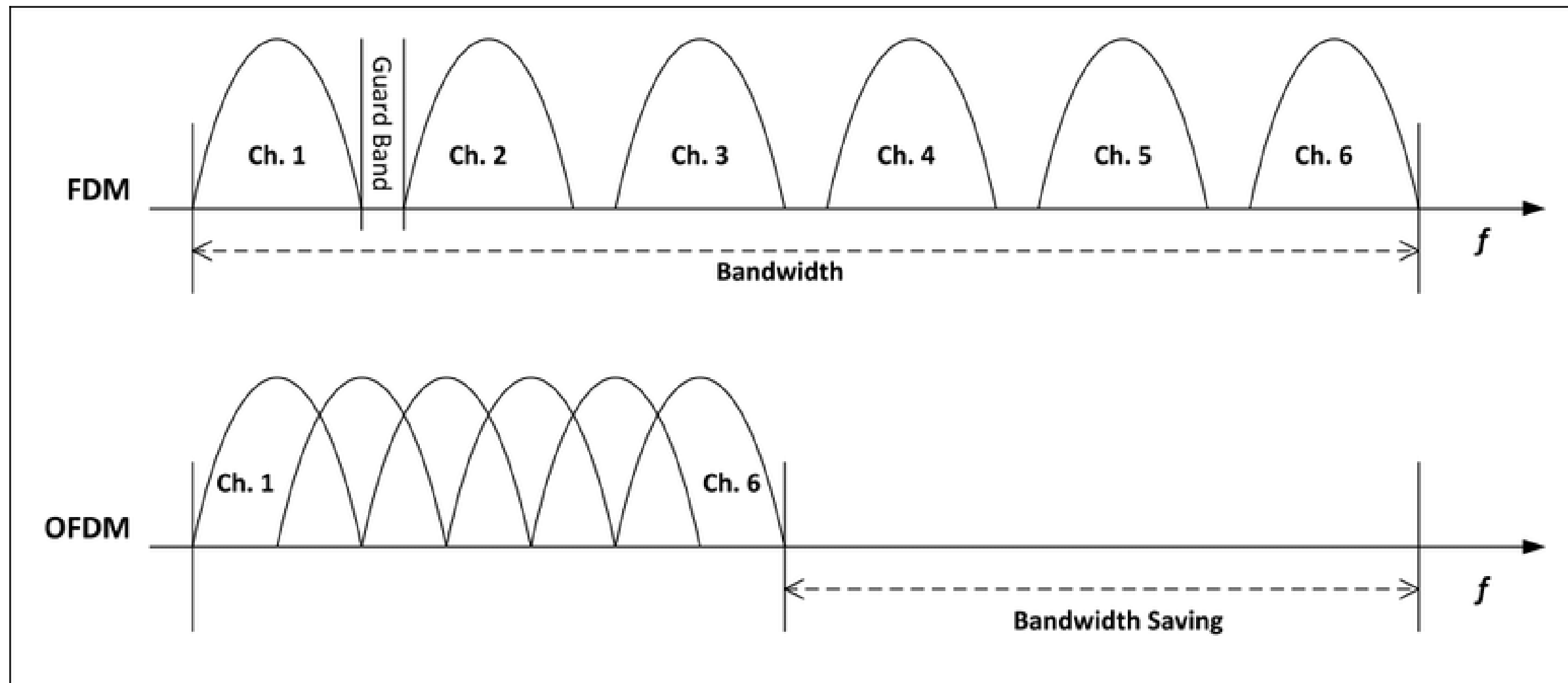
Channel Bonding

- 20 + 20 -> 40MHz
- High throughput



updated OFDM

- Orthogonal Frequency Division Multiplexing
- # of subcarrier for data : 802.11 a,g (48) -> 802.11n (52)



MIMO

- Multiple Input Multiple Output
- Improve reliability
- Increase throughput



III wi-fi 5

Standard	802.11n	802.11ac	802.11ax
Year	2009	2013	2019
Frequency band	2.4GHz, 5GHz	5GHz	2.4GHz, 5GHz
channel bandwidth	20, 40	20, 40, 80, 160	20, 40, 80, 160
Max data rate	600 Mbps	7Gbps	9.6 Gbps
MIMO	MIMO (up to 4X4)	MU-MIMO (DL, up to 8X8)	MU-MIMO (DL&UL)
Modulation	OFDM	OFDM	OFDM, OFDMA
Data subcarrier modulation	BPSK, QPSK, 16QAM, 64QAM	BPSK, QPSK, 16QAM, 64QAM, 256QAM	BPSK, QPSK, 16QAM, 64QAM, 256QAM, 1024QAM

Wi-fi4 vs Wave1



Wi-fi 4

- 64QAM
- 40MHz channel

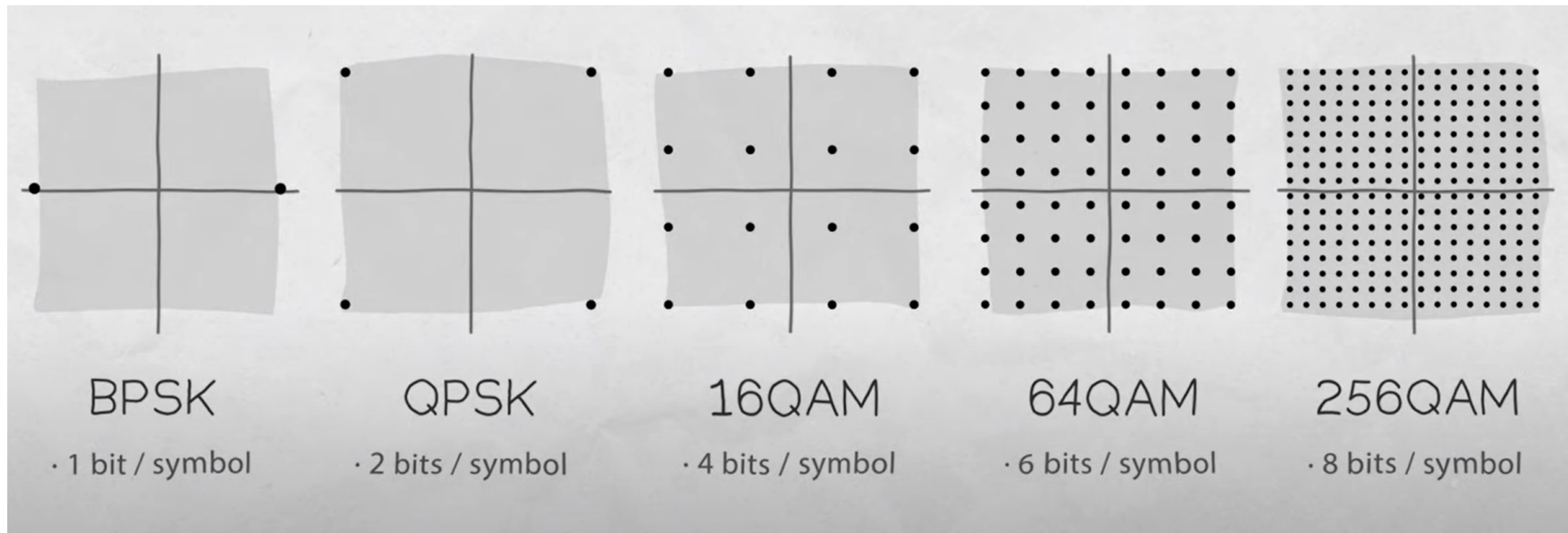


Wave1

- 256QAM
- 80MHz channel

QAM (Quadrature Amplitude Modulation)

- Modulation scheme
- The larger the number of the QAM, the more data can transfer



PSK (Phase Shift Keying)

Wave1 vs Wave2

Wave1

80MHz
channel

MIMO

Wave2

160MHz
channel

MU-MIMO
(DL)

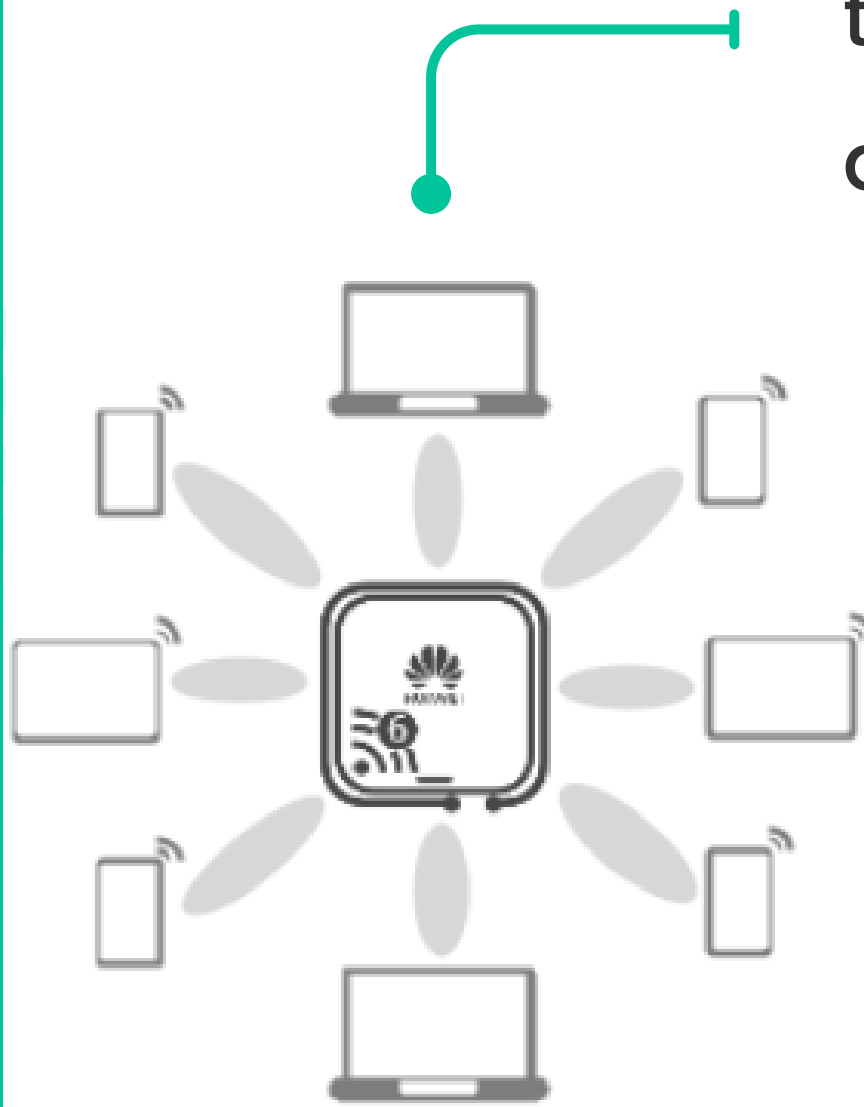
MIMO vs MU-MIMO

only communicate
with **one** connected
device at a time



MU-MIMO

can send information
to **multiple** devices
at the **same time**



only support
Down Link in wifi5

Beamforming

- Strengthen signals in specific directions from AP to client
- Reduce unnecessary RF interference
- Stronger, faster, more reliable wireless communication



(1) Without Beamforming

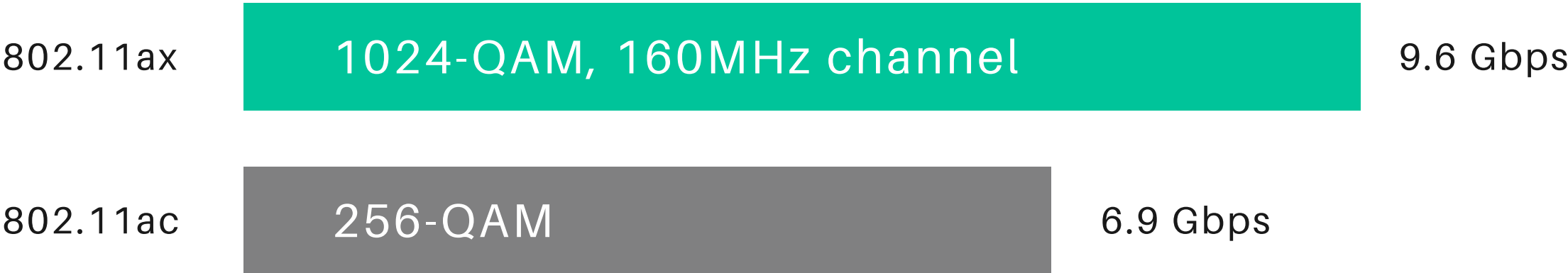


(2) Beamforming

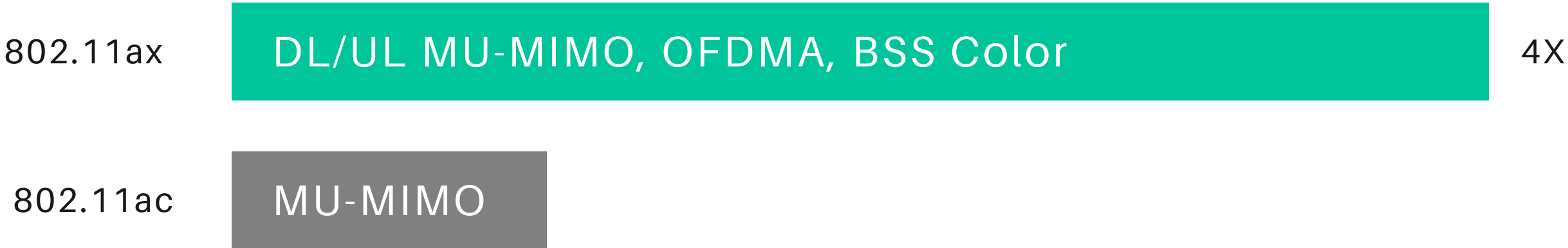
IV wi-fi 6

Standard	802.11n	802.11ac	802.11ax
Year	2009	2013	2019
Frequency band	2.4GHz, 5GHz	5GHz	2.4GHz, 5GHz
channel bandwidth	20, 40	20, 40, 80, 160	20, 40, 80, 160
Max data rate	600 Mbps	7Gbps	9.6 Gbps
MIMO	MIMO (up to 4X4)	MU-MIMO (DL, up to 8X8)	MU-MIMO (DL&UL)
Modulation	OFDM	OFDM	OFDM, OFDMA
Data subcarrier modulation	BPSK, QPSK, 16QAM, 64QAM	BPSK, QPSK, 16QAM, 64QAM, 256QAM	BPSK, QPSK, 16QAM, 64QAM, 256QAM, 1024QAM

Fast Speed



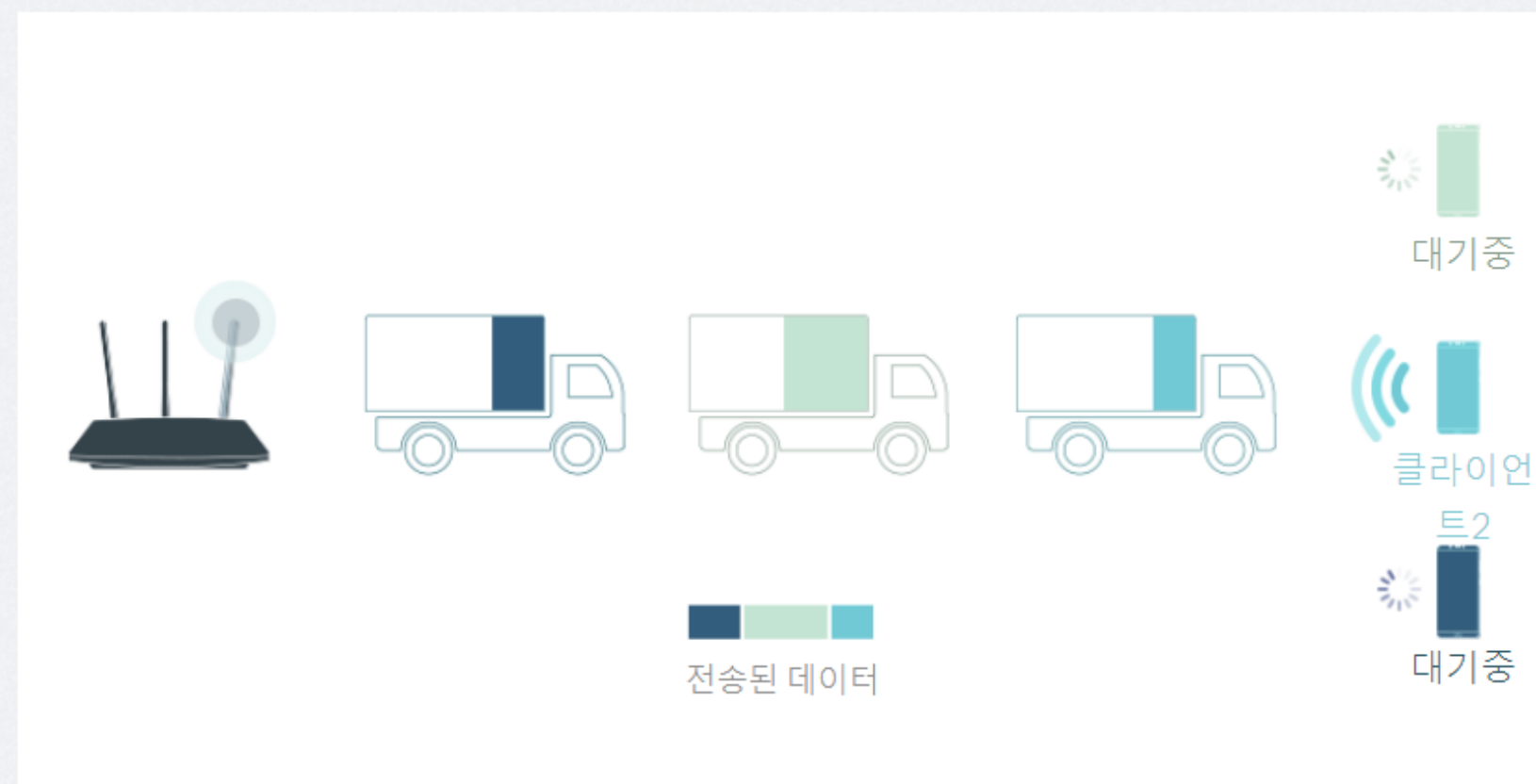
Large Capacity



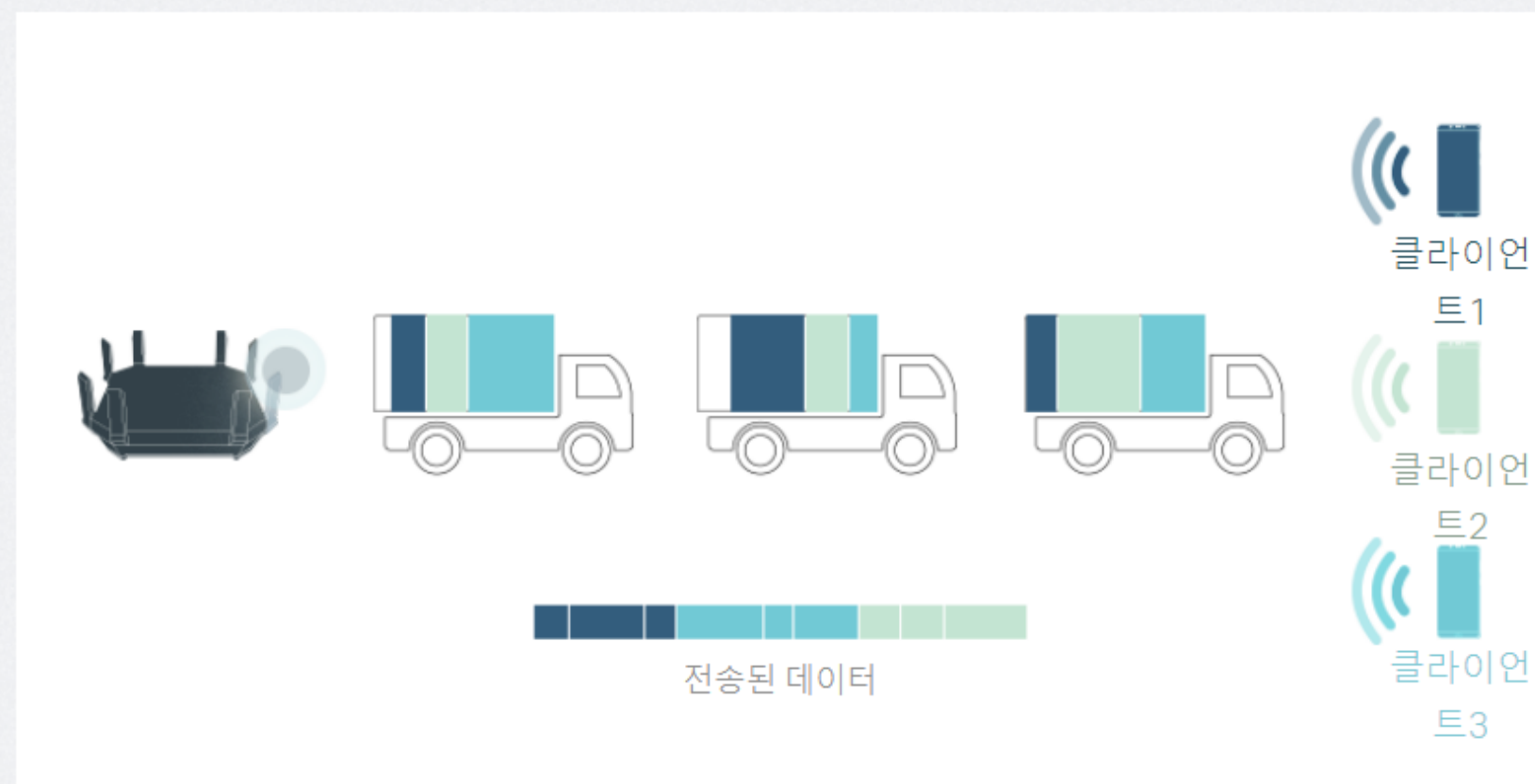
OFDMA

- Multiple user at the same time

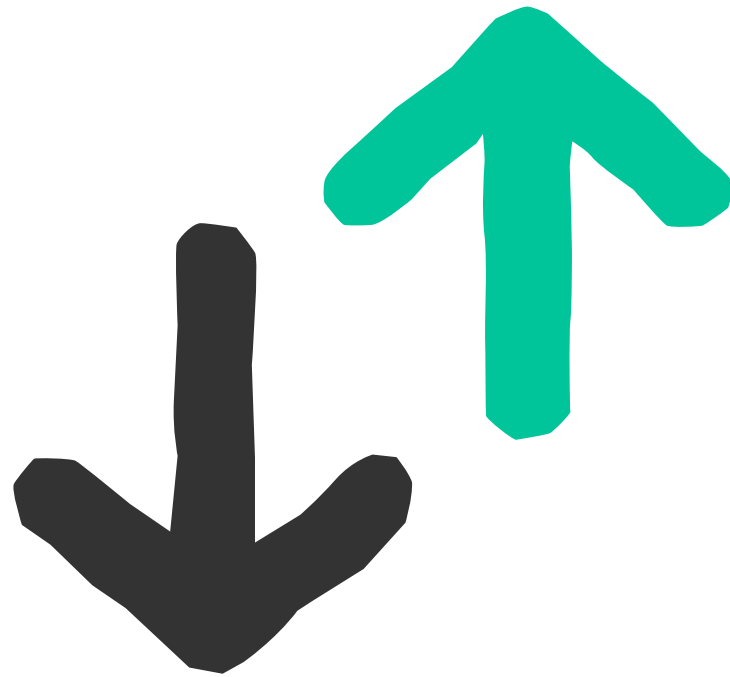
● 아래 그림은 하나의 안테나만 작동하는 상황입니다.



OFDMA이 없는 802.11ac

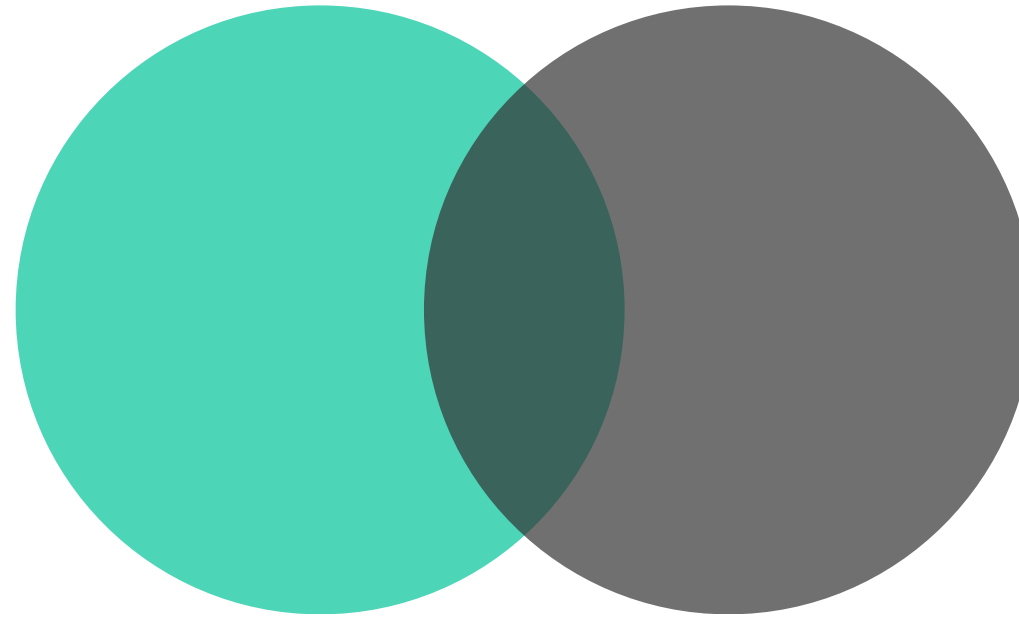


OFDMA이 있는 802.11ax



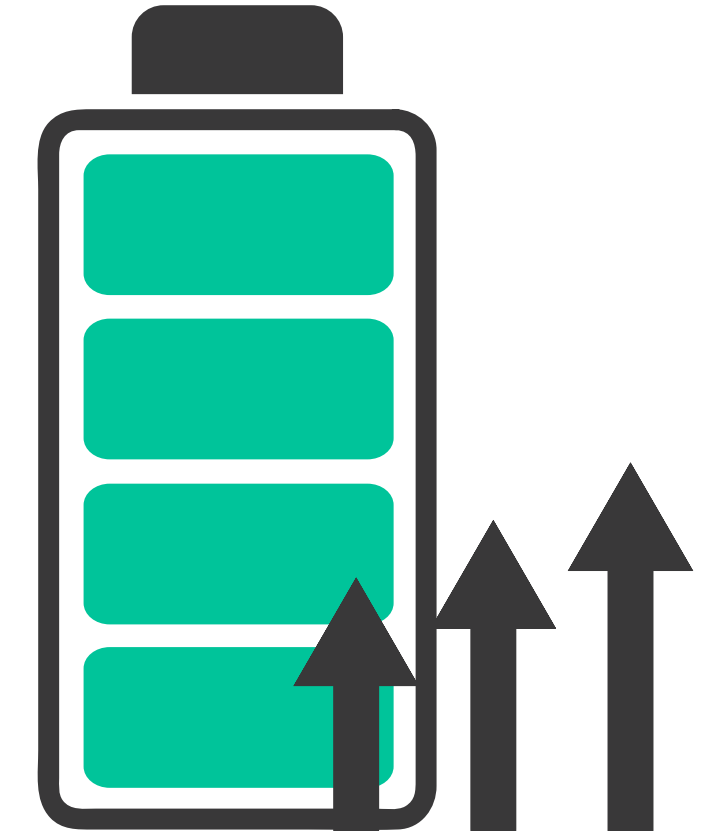
UL MIMO

- both DL & UL
- 8x8 spatial stream



BSS Coloring

- Minimize interference with neighboring Wi-Fi

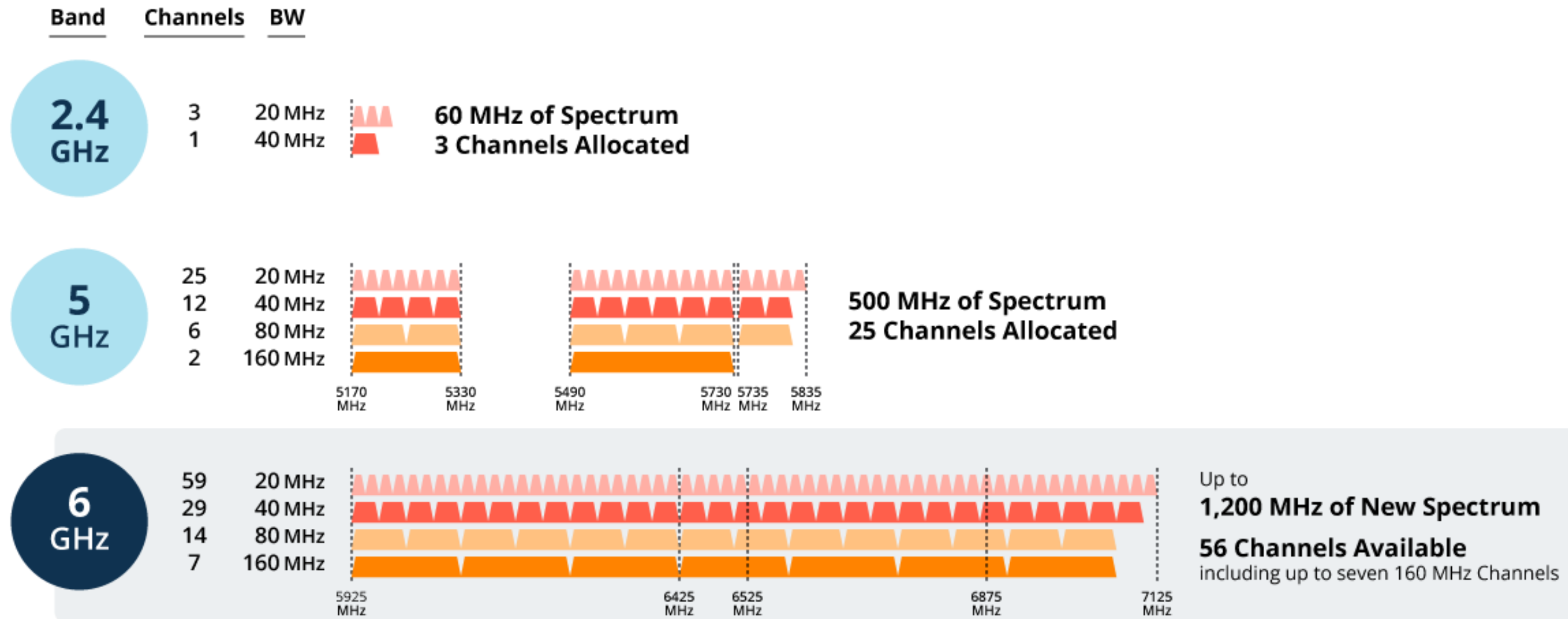


TWT

- Target Wakeup Time
- long battery life

Wi-fi 6E

- 6GHz band



Spectrum available in the 6 GHz band varies by geography.

Q&A